the natural isomorphism $\pi_1(K) \approx H_1(K;\mathbb{Z})$ we know that f induces multiplication by r on $H_1(K;\mathbb{Z})$. Via the universal coefficient theorem, f also induces multiplication by r on $H^1(K;\mathbb{Z}_{p^n})$ and $H^2(K;\mathbb{Z}_{p^n})$. The cup product structure in $H^*(K;\mathbb{Z}_{p^n})$ computed in Examples 3.41 and 3E.2 then implies that f induces multiplication by r^i on $H^{2i-1}(K;\mathbb{Z}_{p^n})$, so the same is true for $H_{2i-1}(K;\mathbb{Z})$ by another application of the universal coefficient theorem.

For each integer $j \geq 0$ let $h_j : \Sigma K \to \Sigma K$ be the difference $\Sigma f - r^j \mathbb{1}$, so h_j induces multiplication by $r^i - r^j$ on $H_{2i}(\Sigma K;\mathbb{Z}) \approx \mathbb{Z}_{p^n}$. By the choice of r we know that p divides $r^i - r^j$ iff $i \equiv j \bmod p-1$. This means that the map induced by h_j on $\widetilde{H}_{2i}(\Sigma K;\mathbb{Z})$ has nontrivial kernel iff $i \equiv j \bmod p-1$. Therefore the composition $m_i = h_1 \circ \cdots h_{i-1} \circ h_{i+1} \cdots h_{p-1}$ induces an isomorphism on $\widetilde{H}_*(\Sigma K;\mathbb{Z})$ in dimensions congruent to $2i \bmod 2p-2$ and has a nontrivial kernel in other dimensions where the homology group is nonzero. When there is a nontrivial kernel, some power of m_i will induce the zero map since we are dealing with homomorphisms $\mathbb{Z}_{p^n} \to \mathbb{Z}_{p^n}$.

Let X_i be the mapping telescope of the sequence $\Sigma K \to \Sigma K \to \cdots$ where each map is m_i. Since homology commutes with direct limits, the inclusion of the first factor $\Sigma K \hookrightarrow X_i$ induces an isomorphism on $\widetilde{H}_*$ in dimensions congruent to $2i \bmod 2p-2$, and $\widetilde{H}_*(X_i;\mathbb{Z}) = 0$ in all other dimensions. The sum of these inclusions is a map $\Sigma K \to X_1 \vee \cdots \vee X_{p-1}$ inducing an isomorphism on all homology groups. Since these complexes are simply-connected, the result follows by Whitehead's theorem. $\square$

The construction of the spaces X_i as mapping telescopes produces rather large spaces, with infinitely many cells in each dimension. However, by Proposition 4C.1 each X_i is homotopy equivalent to a CW complex with the minimum configuration of cells consistent with its homology, namely, a 0-cell and a k-cell for each k congruent to $2i$ or $2i+1 \bmod 2p-2$.

Stable splittings of $K(G,1)$'s for finite groups G have been much studied and are a complicated and subtle business. To take the simplest noncyclic example, Proposition 4I.1 implies that $\Sigma K(\mathbb{Z}_2 \times \mathbb{Z}_2, 1)$ splits as the wedge sum of two copies of $\Sigma K(\mathbb{Z}_2, 1)$ and $\Sigma(K(\mathbb{Z}_2,1) \wedge K(\mathbb{Z}_2,1))$, but the latter summand can be split further, according to a result in [Harris & Kuhn 1988] which says that for G the direct sum of k copies of $\mathbb{Z}_{p^n}$, $\Sigma K(G,1)$ splits canonically as the wedge sum of pieces having exactly $p^k - 1$ distinct homotopy types. Some of these summands occur more than once, as we see in the case of $\mathbb{Z}_2 \times \mathbb{Z}_2$.

Exercises

1. If a connected CW complex X retracts onto a subcomplex A, show that $\Sigma X \simeq \Sigma A \vee \Sigma(X/A)$. [One approach: Show the map $\Sigma r + \Sigma q : \Sigma X \to \Sigma A \vee \Sigma(X/A)$ induces an isomorphism on homology, where $r : X \to A$ is the retraction and $q : X \to X/A$ is the quotient map.]

2. Using the Künneth formula, show that $\Sigma K(\mathbb{Z}_m \times \mathbb{Z}_n, 1) \simeq \Sigma K(\mathbb{Z}_m, 1) \vee \Sigma K(\mathbb{Z}_n, 1)$ if m and n are relatively prime. Thus to determine stable splittings of $K(\mathbb{Z}_n, 1)$ it suffices to do the case that n is a prime power, as in Proposition 4I.3.

3. Extending Proposition 4I.3, show that the $(2k+1)$-skeleton of the suspension of a lens space with fundamental group of order p^n is homotopy equivalent to the wedge sum of the $(2k+1)$-skeleta of the spaces X_i, if these X_i's are chosen to have the minimum number of cells in each dimension, as described in the remarks following the proof.

4.J The Loopspace of a Suspension

Loopspaces appear at first glance to be hopelessly complicated objects, but if one is only interested in homotopy type, there are many cases when great simplifications are possible. One of the nicest of these cases is the loopspace of a sphere. We show in this section that ΩS^{n+1} has the weak homotopy type of the James reduced product $J(S^n)$ introduced in §3.2. More generally, we show that $\Omega\Sigma X$ has the weak homotopy type of $J(X)$ for every connected CW complex X. If one wants, one can strengthen 'weak homotopy type' to 'homotopy type' by quoting Milnor's theorem, mentioned in §4.3, that the loopspace of a CW complex has the homotopy type of a CW complex.

Part of the interest in $\Omega\Sigma X$ can be attributed to its close connection with the suspension homomorphism $\pi_i(X) \to \pi_{i+1}(\Sigma X)$. We will use the weak homotopy equivalence of $\Omega\Sigma X$ with $J(X)$ to give another proof that the suspension homomorphism is an isomorphism in dimensions up to approximately double the connectivity of X. In addition, we will obtain an exact sequence that measures the failure of the suspension map to be an isomorphism in dimensions between double and triple the connectivity of X. An easy application of this, together with results proved elsewhere in the book, will be to compute $\pi_{n+1}(S^n)$ and $\pi_{n+2}(S^n)$ for all n.

As a rough first approximation to ΩS^{n+1} there is a natural inclusion of S^n into ΩS^{n+1} obtained by regarding S^{n+1} as the reduced suspension ΣS^n, the quotient $(S^n \times I)/(S^n \times \partial I \cup \{e\} \times I)$ where e is the basepoint of S^n, then associating to each point $x \in S^n$ the loop $\lambda(x)$ in ΣS^n given by $t \mapsto (x,t)$. The figure shows what a few such loops look like. However, we cannot expect this inclusion $S^n \hookrightarrow \Omega S^{n+1}$ to be a homotopy equivalence since ΩS^{n+1} is an H-space but S^n is only an H-space when $n = 1, 3, 7$ by the theorem of Adams discussed in §4.B. The simplest way to correct this deficiency in S^n would be to replace it by the free H-space that it generates, the reduced product $J(S^n)$. Re-